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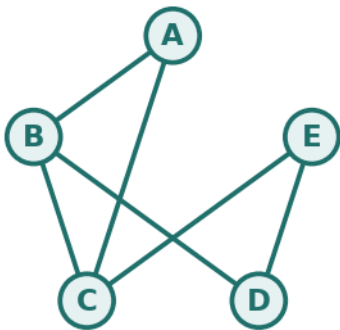
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Learning Objectives

- Compute vertex degrees and apply the Handshaking Theorem
- Identify Euler circuits/paths and Hamiltonian paths
- Write adjacency matrices and interpret graph structure
- Apply Euler's formula $V - E + F = 2$ for planar graphs

For every graph problem: first find all vertex degrees. For Euler circuits: check that ALL vertices have even degree.

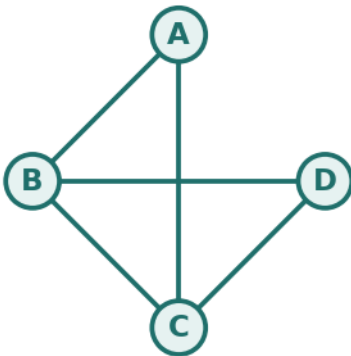
1. Find the degree of each vertex and verify the Handshaking Theorem (sum of degrees = $2|E|$).



Find $\deg(A)$, $\deg(B)$, $\deg(C)$, $\deg(D)$, $\deg(E)$

Answer: _____

2. Does the graph have an Euler circuit? An Euler path? State the condition and check.

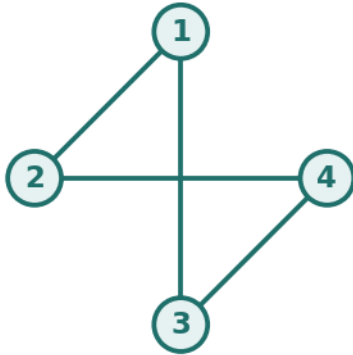


Does an Euler circuit exist?

Answer: _____



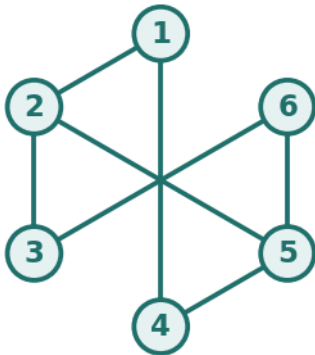
3. Write the adjacency matrix for the graph with vertices $\{1, 2, 3, 4\}$ and edges $\{(1,2),(1,3),(2,4),(3,4)\}$.



Write the 4×4 adjacency matrix.

Answer: _____

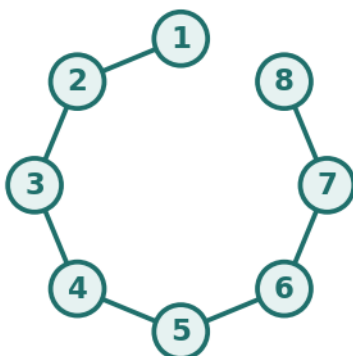
4. Is the graph bipartite? If yes, find the two vertex groups.



Color vertices using 2 colors — bipartite?

Answer: _____

5. A connected graph has 8 vertices and is a tree. How many edges does it have? Draw a possible spanning tree.

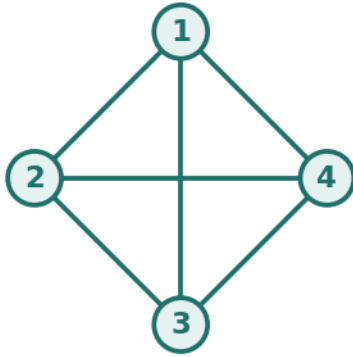


Tree with 8 vertices — how many edges?

Answer: _____



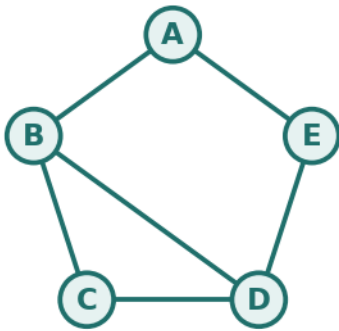
6. What is the chromatic number $\chi(G)$ of a complete graph K_n ? Color it using the minimum number of colors.



K_4 — complete graph on 4 vertices

Answer: _____

7. Find a Hamiltonian path in the graph (a path that visits every vertex exactly once). Does a Hamiltonian circuit exist?



Find a Hamiltonian path.

Answer: _____

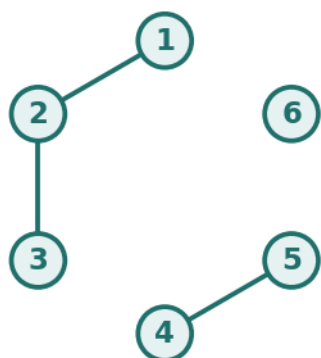
8. A weighted graph has edges: A–B(3), A–C(5), B–C(1), B–D(4), C–D(2). Find the shortest path from A to D.

$$A \xrightarrow{3} B \xrightarrow{4} D, \quad A \xrightarrow{3} B \xrightarrow{1} C \xrightarrow{2} D$$

Answer: _____



9. How many connected components does the graph have? List the vertices in each component.



How many connected components?

Answer: _____

10. A connected planar graph has 10 vertices and 15 edges. How many faces does it have? (Use Euler's formula: $V - E + F = 2$)

$$V - E + F = 2, \quad V = 10, \quad E = 15$$

Answer: _____

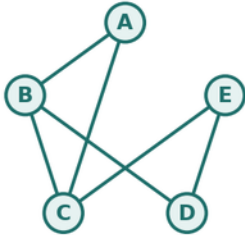




Problems 1–2: degree sum and Euler circuit condition — foundational. Problem 3 (adjacency matrix): connect to matrix concepts from Linear Algebra. Problem 10 (Euler's formula): make sure students count the outer face.

Solutions

- Find the degree of each vertex and verify the Handshaking Theorem (sum of degrees = $2|E|$).

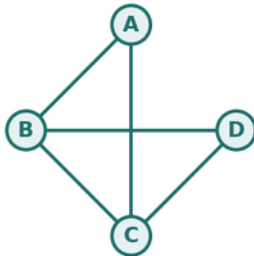


Find $\deg(A)$, $\deg(B)$, $\deg(C)$, $\deg(D)$, $\deg(E)$

- $\deg(A)=2$, $\deg(B)=3$, $\deg(C)=3$, $\deg(D)=2$, $\deg(E)=2$.
- $\text{Sum} = 2+3+3+2+2 = 12$.
- Edges: $(A,B), (A,C), (B,C), (B,D), (C,D), (D,E) \rightarrow |E|=6$.
- Handshaking: $\text{sum of degrees} = 2|E| = 2 \cdot 6 = 12$. ✓

Answer: $\sum \deg = 12 = 2 \times 6$

- Does the graph have an Euler circuit? An Euler path? State the condition and check.



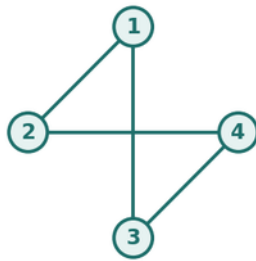
Does an Euler circuit exist?

- Euler circuit exists iff ALL vertices have even degree.
- $\deg(A)=2$, $\deg(B)=3$, $\deg(C)=3$, $\deg(D)=2$.
- B and C have odd degree → no Euler circuit.
- Euler path exists iff exactly 0 or 2 vertices have odd degree — here exactly 2 (B and C). ✓
- Euler path starts at B and ends at C (or vice versa).

Answer: No Euler circuit (odd-degree vertices); Euler path exists



3. Write the adjacency matrix for the graph with vertices $\{1, 2, 3, 4\}$ and edges $\{(1,2),(1,3),(2,4),(3,4)\}$.



Write the 4×4 adjacency matrix.

→ Row/col order: 1,2,3,4.

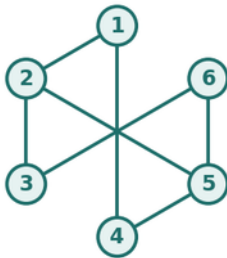
→ Row 1: $(1,1)=0, (1,2)=1, (1,3)=1, (1,4)=0$.

→ Row 2: 1,0,0,1. Row 3: 1,0,0,1. Row 4: 0,1,1,0.

→ Adjacency matrix is symmetric for undirected graphs.

Answer: $A_{ij} = 1$ if $(i, j) \in E$, 0 otherwise

4. Is the graph bipartite? If yes, find the two vertex groups.



Color vertices using 2 colors — bipartite?

→ A graph is bipartite iff it contains NO odd-length cycle.

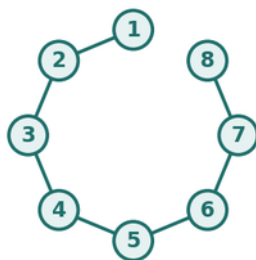
→ Try 2-coloring starting from vertex 1: color alternating.

→ Look for any triangle (3-cycle) or odd cycle — if found, NOT bipartite.

→ With 6 vertices and these edges, trace cycles to check.

Answer: Check: odd cycle $\Rightarrow$ not bipartite

5. A connected graph has 8 vertices and is a tree. How many edges does it have? Draw a possible spanning tree.



Tree with 8 vertices — how many edges?

→ A tree with n vertices has exactly $n-1$ edges.

→ 8 vertices $\rightarrow$ 7 edges.

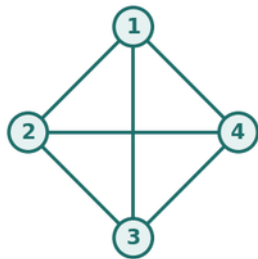
→ A tree is connected and acyclic (no cycles).

→ The drawing shows a path graph — one example of a tree on 8 vertices.

Answer: $|E| = n - 1 = 8 - 1 = 7$ edges



6. What is the chromatic number $\chi(G)$ of a complete graph K_n ? Color it using the minimum number of colors.

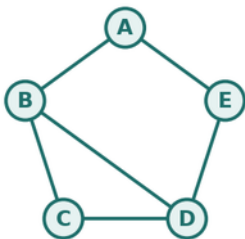


K_4 — complete graph on 4 vertices

- K_n : every vertex is adjacent to every other vertex.
- No two adjacent vertices can share a color.
- All 4 vertices are mutually adjacent → need 4 different colors.
- $\chi(K_n) = n$ for complete graphs.

Answer: $\chi(K_4) = 4$

7. Find a Hamiltonian path in the graph (a path that visits every vertex exactly once). Does a Hamiltonian circuit exist?



Find a Hamiltonian path.

- Hamiltonian path: visit each vertex exactly once.
- Try: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ — check if all edges exist: $(A,B) \checkmark, (B,C) \checkmark, (C,D) \checkmark, (D,E) \checkmark$.
- Hamiltonian circuit: also need edge $(E,A) \rightarrow (E,A) = (4,0) \checkmark$.
- Hamiltonian circuit: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$.

Answer: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$ (circuit exists)

8. A weighted graph has edges: $A-B(3)$, $A-C(5)$, $B-C(1)$, $B-D(4)$, $C-D(2)$. Find the shortest path from A to D.

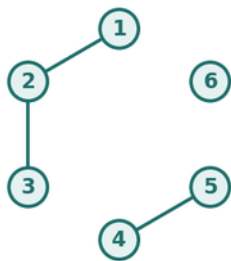
$$A \xrightarrow{3} B \xrightarrow{4} D, \quad A \xrightarrow{3} B \xrightarrow{1} C \xrightarrow{2} D$$

- Option 1: $A \rightarrow B \rightarrow D = 3+4 = 7$.
- Option 2: $A \rightarrow C \rightarrow D = 5+2 = 7$.
- Option 3: $A \rightarrow B \rightarrow C \rightarrow D = 3+1+2 = 6$. ← shortest
- Shortest path: $A \rightarrow B \rightarrow C \rightarrow D$ with total weight 6.

Answer: $A \rightarrow B \rightarrow C \rightarrow D$, cost = $3 + 1 + 2 = 6$



9. How many connected components does the graph have? List the vertices in each component.



How many connected components?

→ Component 1: vertices 1–2–3 connected via edges (1,2),(2,3).

→ Component 2: vertices 4–5 connected via edge (4,5).

→ Component 3: vertex 6 is isolated (no edges).

→ 3 connected components total.

Answer: 3 components: {1, 2, 3}, {4, 5}, {6}

10. A connected planar graph has 10 vertices and 15 edges. How many faces does it have? (Use Euler's formula: $V - E + F = 2$)

$$V - E + F = 2, \quad V = 10, \quad E = 15$$

→ Euler's formula for connected planar graphs: $V - E + F = 2$.

$$\rightarrow F = 2 - V + E = 2 - 10 + 15 = 7.$$

→ One of the 7 faces is the outer (unbounded) face.

Answer: $F = 2 - V + E = 2 - 10 + 15 = 7$ faces

